PYRA

Forest-Floor Wildfire Intelligence Platform

Project Brief • Version 0.1 • Initial Brief for Development Agent

1. Mission

PYRA is a ruggedized, stake-shaped sensor node driven into forest ground to continuously measure the physical conditions that govern wildfire ignition and spread. Each node streams data to a cloud platform that fuses on-device sensing with satellite, terrain, weather, and vegetation datasets to produce live fire-danger intelligence for two audiences:

1. **Operational:** firefighters, fire agencies, incident commanders, and dispatchers who need localized, real-time situational awareness.
2. **Scientific / managerial:** forest managers, ecologists, and researchers who need long-term data on fuel conditions, forest dynamics, and climate change impacts on fire regimes.

> PYRA is **not** a new fire-danger model. It is a sensing-and-data platform that feeds established fire-science frameworks (Canadian FWI/FBP, US NFDRS2016, Nelson fuel-moisture models) with better, denser, ground-truth data than any current system.

2. Design Principles

3. **Measure, don't invent.** Whenever an established fire-science model exists, we compute it — we do not replace it.
4. **Sensor fusion over single signals.** No single sensor tells the truth; combinations do.
5. **Graceful degradation.** A node that loses comms should still store data locally. A node that loses power should still flash a last-gasp alert. A network that loses half its nodes should still be useful.
6. **Dual-use from day one.** The same raw data feeds firefighter dashboards and research pipelines — different consumers, same source of truth.
7. **Open interfaces.** Use open standards (OGC SensorThings, STAC, GeoJSON, MQTT) so PYRA data can flow into existing agency GIS tools.
8. **Never create false confidence.** Every derived value ships with an uncertainty estimate. Firefighters make life-and-death decisions; they must be able to trust — and appropriately distrust — our outputs.
9. **Expendable where necessary.** Nodes in a fire's path will be lost. Design the unit cost and the data pipeline around that reality.

3. System Architecture

A one-page view of the platform, from forest floor to consumer app:

```
FIELD LAYER
  PYRA Node  ─┐
  PYRA Node  ─┼── LoRaWAN / BLE mesh ──▶  PYRA Gateway ──▶  encrypted uplink
  PYRA Node  ─┘                            (LoRaWAN + cellular/satellite)

CLOUD LAYER
  Ingest  ──▶  Time-series DB (Timescale)  ──▶  GIS / PostGIS
                                                  + Fire-Science Engine
                                                    (FWI, FBP, Nelson, NFDRS)
  Alerting  ◀──┘           Fleet Mgmt             OTA
  External feeds: Sentinel-2, DEM, NOAA/ECCC, FIRMS, OSM, CNFDB, LANDFIRE,
                  lightning networks

CONSUMER LAYER
  Firefighter mobile app (iOS/Android)
  Research / management web dashboard
  Agency GIS plug-ins  (ArcGIS / QGIS)
  Public API + data catalogue
```

4. Hardware — The PYRA Node

4.1 Form factor

- **Stake geometry.** ~15 cm conical stainless-steel ground spike, ~40 cm above-ground tube housing electronics, ~25 cm mast for wind sensor / solar panel. Total height ~80 cm deployed.
- **Ground coupling.** Soil-contact sensors on the buried spike; above-ground sensors in the tube; wind and sun on the mast.
- **Enclosure.** IP67 minimum, UV-stable polymer + stainless hardware, powder-coated safety-orange for visibility to ground crews. Ceramic-lined upper cap to extend heat survival. Will not survive direct crown fire — don't pretend otherwise.
- **Deployment methods:** hand-deployed (primary), drone-dropped, vehicle- or helicopter-launched for rapid pre-positioning.

4.2 Core electronics

Component	Suggested part	Rationale
MCU	ESP32-S3 + LoRa SX1262 companion, or single-chip nRF9160 for cellular	Mature toolchain, cheap, strong community
Storage	8–32 MB flash + microSD slot	Local buffer for days of data during comms outage
GNSS	u-blox MAX-M10S	Low-power, multi-constellation
RTC	DS3231 w/ coin cell	Accurate timestamps during sleep

Component	Suggested part	Rationale
Security	ATECC608B secure element	Key storage, device attestation, tamper evidence

4.3 Sensor suite

Parameter	Sensor / technique	Notes
Soil moisture (2 depths)	Capacitive probe (Teros 10 class)	Proxy for drought stress; calibrate per soil type
Soil temperature (2 depths)	DS18B20 on buried cable	Pairs with moisture for energy balance
Duff moisture (surface)	Fuel-stick replica + capacitive read	Approximates FFMC inputs
Air temperature + RH	SHT45	Core input to FWI and fuel-moisture models
Barometric pressure	BMP390	Weather context, altitude check
Wind speed + direction	Ultrasonic anemometer (FT205)	No moving parts; critical for ISI / FBP
Solar irradiance	Silicon pyranometer (Apogee SP-110)	Drives fuel drying, evapotranspiration
UV index	SI1133 / LTR390	Cheap supplement to pyranometer
Rain / leaf wetness	Tipping-bucket + capacitive pad	Rainfall is the dominant FFMC driver
Smoke / particulates (PM2.5/10)	Plantower PMS7003 or Sensirion SPS30	Early fire signal; air-quality value
Gas: CO, CO_2, VOC	MQ-series + CCS811 / SGP41	High false-positive — fuse with PM + thermal
Thermopile / IR hotspot	Melexis MLX90640 (32×24 IR array)	Nearby heat anomaly + canopy temp
Lightning	AS3935 Franklin detector	Ignition-cause tracking; storm warning
Acoustic	MEMS mic + TinyML classifier	Chainsaw, vehicle, gunshot, fire-crackle, wildlife
PIR motion	Low-power passive IR	Large-mammal + human presence (fuse with acoustic)
Accelerometer + magnetometer	ICM-20948	Slope/aspect at install, tamper/knockdown

Parameter	Sensor / technique	Notes
Camera (higher SKU)	OV5640 + Himax WE-I Plus NPU	Fuel density, ladder fuels, snow, canopy closure

Not everything ships in every SKU — see §10 Product Tiers.

4.4 Power

- **Solar:** 5–10 W panel, MPPT charge controller (CN3791 or similar).
- **Battery:** $LiFePO_4$ 18650 pack — safer thermal behaviour than Li-ion in forest environments, wider temperature range.
- **Budget target:** survive 14 consecutive days of full canopy shade / cloud cover.
- **Power modes:** deep sleep default; wake on schedule + interrupt (lightning, accelerometer shock, thermopile threshold, rain start).
- **Action item:** measure average current draw per subsystem empirically and build a power-budget spreadsheet before designing the PCB.

4.5 Deployment-side niceties

- **QR code** on each node, tying physical unit to cloud record.
- **NFC tag** for firefighter to tap with phone and pull last 24 h of data offline.
- **E-ink display** (1.5") confirms successful install: node ID, GPS lock, radio signal, battery.
- **Hall-effect "arm" sensor:** magnet on a lanyard activates the node; eliminates shelf/shipping drain.
- **Capacitive touch / button** for manual alert or self-test.

5. Communications

5.1 Decision matrix

Link	Range	Power	Bandwidth	Good for
BLE mesh (node ↔ node)	~50 m	Very low	Low	Close-in redundancy, firefighter phone link
LoRaWAN	2–15 km LOS	Low	Very low	Primary backhaul in populated-ish forest
LTE-M / NB-IoT	Carrier-dep.	Moderate	Low-mid	Where cellular exists
Satellite (Swarm, Iridium SBD, Astrocast)	Global	Moderate	Very low	Backcountry; the killer feature for firefighting

Link	Range	Power	Bandwidth	Good for
Wi-Fi (gateway only)	50–200 m	High	High	Gateway uplink / maintenance

5.2 Topology

- **Node → Gateway → Internet** is the primary path (LoRaWAN).
- **Gateway** sits in a tower, ridge, or mobile command post. One gateway supports 100+ nodes in a 5–10 km radius.
- **Satellite-equipped super-nodes:** every Nth node carries satellite backup so isolated clusters still report critical alerts.
- **BLE local-dump fallback:** firefighter walking near any node pulls buffered data via phone even with no network.

5.3 Protocol & payload

- **Uplink:** CBOR-encoded payloads over MQTT (via ChirpStack LoRaWAN NS) or MQTT-SN for cellular.
- **Schema:** publish a versioned schema; include schema version in every payload.
- **Three message classes:** Heartbeat (lossy OK); Telemetry (retry with backoff); Alert (immediate, duplicated across radios).

5.4 Security

- TLS on every hop that supports it.
- ATECC608B device keys; mutual auth with network server.
- Signed OTA updates; mandatory rollback slot.
- Rate-limit alerts to prevent a compromised node from DoS-ing the dispatch channel.

6. Software Stack

6.1 Firmware (node)

- **Language:** C/C++ via ESP-IDF or Zephyr RTOS (Zephyr for MCU portability).
- **Modules:** sensor drivers, sensor-fusion layer, power manager, comms stack, local alert engine, OTA agent, log store.
- **Edge ML:** TFLite-Micro for acoustic classifier and optional vision. Models under 512 KB, int8 quantized.
- **Local alert engine:** runs simple thresholds with no network — e.g. thermopile > 80 °C AND PM2.5 rising AND RH falling → local alarm + store-and-forward.

6.2 Cloud backend

- **Ingest:** MQTT broker (EMQX / Mosquitto) behind a TLS-terminating proxy.

- **Stream processor:** lightweight Rust/Go worker that validates, decodes, and routes.
- **Storage:** TimescaleDB (SQL + time-series + PostGIS in one) or InfluxDB; S3-compatible blob store for images/audio/firmware.
- **Fire-science engine:** scheduled + event-driven worker computing FWI/FBP/NFDRS/Nelson per node on each update, with provenance.
- **External data pipeline:** daily/weekly ETL for satellite, weather, lightning, OSM — cached and attached to each node's spatial record.
- **Alert engine:** rule DSL writable by agency users; escalation tiers (watch/warning/emergency); pluggable notifications (SMS, email, Slack/Teams, radio-bridge tone, mobile push).
- **Fleet management:** inventory, health dashboard, OTA rollout control, recall/decommission workflow.

6.3 Consumer apps

- **Firefighter mobile app** (React Native): map-first, offline tiles, node status + derived index, install workflow, BLE local-dump, incident overlay with wind + slope + predicted spread.
- **Research / management web app** (React + MapLibre): historical query, CSV/GeoJSON/NetCDF export, model validation tools.
- **Agency GIS plug-ins:** ArcGIS Pro toolbox and QGIS plug-in consuming the SensorThings API.
- **Public/open data portal:** subset released under an appropriate licence (see §13).

6.4 APIs

- **OGC SensorThings API** for sensor observations (standard in environmental monitoring).
- **STAC** catalogue for satellite/raster derivatives.
- **REST + GeoJSON** for map clients.
- **Webhooks** for alert subscribers.
- **IRWIN adapter** (Integrated Reporting of Wildland-Fire Information) for US fire agencies.

7. External Data Integrations

Build adapters, not one-offs. Each integration should be a pluggable module with caching, rate-limiting, and provenance logging.

Dataset	Source	Use
Digital Elevation Model	USGS 3DEP (US) / CDEM (Canada) / Copernicus GLO-30	Slope, aspect, sun exposure, spread modeling
Sentinel-2 / Landsat	Copernicus / USGS	NDVI, EVI, NBR, NDMI (live fuel moisture proxy)
MODIS / VIIRS FIRMS	NASA	Nearby active fires
Hansen Global Forest	UMD / Google	Forest age, disturbance history

Dataset	Source	Use
Change		
LANDFIRE / Canadian NFIS	USGS / NRCan	Fuel-type classification
Fire-perimeter history	NIFC, CNFDB, EFFIS	Burn history, repeat-burn risk
Weather forecast	NWS / ECCC GeoMet / Open-Meteo	Derived fire-danger forecasting
Lightning	Vaisala NLDN / CLDN / Earth Networks	Ignition cause + storm warning
Snow cover	MODIS MOD10 / Sentinel-1 SAR	Start of fire season
Roads / rivers / structures	OpenStreetMap / cadastre	Natural + man-made barriers, access
Drought indices	NOAA CPC / Canadian Drought Monitor	Regional context
Air quality	AirNow / ECCC AQHI	Context for PM readings

8. Fire-Science Frameworks (compute, don't invent)

These are the models PYRA data feeds into. Implement each as a versioned library module with tests against published reference cases.

- **Canadian Forest Fire Weather Index (FWI) System** — FFMC, DMC, DC, ISI, BUI, FWI. *Especially important in Quebec; aligns with SOPFEU operations.*
- **Canadian Fire Behaviour Prediction (FBP) System** — FWI + fuel type + slope → rate of spread, fire intensity, crown fire likelihood.
- **US NFDRS2016** — cross-border compatibility.
- **Nelson dead-fuel moisture model** — 1-hr, 10-hr, 100-hr, 1000-hr fuel moistures from weather time series.
- **Fosberg Fire Weather Index** — instantaneous, widely understood.
- **Haines Index** — atmospheric instability contribution to plume-dominated fires.
- **KBDI (Keetch-Byram Drought Index)** — cumulative drought proxy.

Every output carries: model name + version (e.g. `fwi_cfs_2019`); input provenance (sensors + feeds that contributed); uncertainty estimate where derivable.

9. What the Node Cannot Sense (and How We Fill It)

A buried stake cannot directly measure several items on the original wish-list. Here's how we get them anyway:

Wish-list item	How we get it
Forest age	Hansen Global Forest Change + national forest inventory (per-pixel lookup)
Forest density / canopy closure	Sentinel-2 NDVI/LAI + LiDAR where available + install-time photo
Species / type of trees	LANDFIRE / NFIS classification + optional install-time tagging
Ladder fuels, dead-and-down	Install-time photo + ML classifier in deployment app
Human presence	Acoustic (chainsaw/vehicle) + PIR + proximity to OSM roads/trails
Natural barriers	OSM + hydrography layers
Snow state	MODIS snow cover + node air temp + optional camera
"Old bush with dead interior"	Install-time photo → vision model scoring fuel-load severity

> The **install-time app workflow** is therefore a core product surface, not a nice-to-have.

10. Product Tiers (SKUs)

Three tiers let agencies mix-and-match based on strategic value of each site:

Tier	What's in it	Target price
PYRA Scout	Core weather + soil + smoke + thermopile + LoRaWAN	Low (dense deployment, semi-expendable)
PYRA Sentinel	Scout + anemometer + lightning + gas suite + satellite backup	Mid
PYRA Vista	Sentinel + camera + edge ML + higher-res IR array	High (hero nodes at strategic points)

Gateways are a separate SKU: **PYRA Relay** (solar-powered LoRaWAN gateway with cellular + satellite uplink).

11. MVP Roadmap

Phase 0 — Paper & bench (weeks 0–6)

Power-budget spreadsheet. Sensor BOM with two candidate parts per slot. Comms decision locked per target geography. Regulatory gap analysis.

Phase 1 — Single-node proof (weeks 6–16)

- One hand-soldered node: ESP32-S3 + SHT45 + BMP390 + capacitive soil probe + ultrasonic anemometer + PMS7003 + LoRa.
- ChirpStack + TimescaleDB + PostGIS on a single VM.
- Canadian FWI computed and graphed over a week of data.

- Web map showing the one stake.

Phase 2 — Small fleet + external data (weeks 16–28)

- 10 nodes + 1 gateway in a pilot forest (partner with SOPFEU or a university research forest).
- DEM + Sentinel-2 + weather forecast integrations live.
- OGC SensorThings API published.
- Install-time mobile app (basic: photo, GPS, fuel tagging, commissioning check).

Phase 3 — Operational pilot (weeks 28–52)

- Satellite-backed super-nodes.
- Alerting with escalation rules.
- Firefighter mobile app v1 with offline maps.
- OTA update pipeline.
- First operational trial with a fire agency partner.

Phase 4 — Scale (year 2)

- Edge ML (vision + acoustic).
- Drone-droppable form factor.
- Research-grade data products (monthly QA/QC'd open datasets).
- Integration with IRWIN, ArcGIS, etc.

12. Risks & Mitigations

Risk	Mitigation
False positives erode firefighter trust	Multi-sensor fusion; tiered alerts; uncertainty estimates; human-in-the-loop for operational alerts
Sensor drift (esp. soil moisture, gas)	Seasonal calibration protocol; anomaly detection vs. neighbour nodes
Node destruction in fires	Design unit economics assuming % loss per season; cheap Scout tier; satellite super-nodes survive longer
Wildlife / vandalism / theft	Tamper detection via accelerometer; safety-orange + "Scientific Equipment" label; GPS theft alerts
Comms blackout at the critical moment	BLE local dump; satellite backup on super-nodes; store-and-forward
Privacy (cameras, acoustic, humans)	On-device classification only; no raw audio/video leaves the node; privacy-by-design review; clear public policy
Data sovereignty (Canadian / provincial)	Canadian-hosted cloud option (AWS ca-central-1 / Azure Canada / OVH Beauharnois); ISED compliance audit

Risk	Mitigation
Regulatory delays on radio certification	Use pre-certified LoRa/cellular modules; budget for FCC/ISED testing early
Firefighter trusts PYRA alone — bad outcomes	UX + training position PYRA as decision support, never sole authority; always display uncertainty

13. Regulatory, Standards & Compliance

- **Radio certification:** FCC Part 15 (US), ISED RSS-247/210 (Canada), CE RED (EU).
- **Safety-of-use near firefighters:** review against NFPA standards (e.g. 1977 PPE-compatibility of markings, 1500 programme compatibility) and CAN/ULC equivalents.
- **Battery transport:** UN 38.3 for $LiFePO_4$ cells.
- **Data standards:** OGC SensorThings, OGC API – Features, STAC, NetCDF-CF for research export.
- **Open licences:** CC-BY-4.0 for public-facing derived datasets; separate commercial licence for real-time feeds.
- **Privacy:** PIPEDA (Canada), Quebec Law 25, and equivalents per deployment region. Data-protection impact assessments before any camera/acoustic SKU ships.

14. Operational & Lifecycle Considerations

- **Deployment planning.** GIS tool suggesting stake locations from terrain, fuel type, and access — avoid dense clustering; target ridge-lines, saddles, fuel-type boundaries, historical ignition hotspots.
- **Field maintenance.** Batteries + desiccants replaceable without tools; OTA firmware; mobile-app-driven calibration routines.
- **End-of-life.** Return-and-recycle programme; lithium cells through partner; polymer selected for recyclability.
- **Training.** Half-day deployment course with e-learning + printed field card; materials version-controlled alongside firmware.
- **Chain of custody.** Every observation signed with device key; immutable audit log (important for post-fire investigations).

15. Integration Targets

- **SOPFEU** (Société de protection des forêts contre le feu — Quebec) — natural first partner.
- **Parks Canada** and provincial fire agencies.
- **USFS / CAL FIRE / NIFC / IRWIN** (US).
- **ALERTCalifornia / ALERTWildfire** camera networks — overlay PYRA on the same maps.
- **WIFIRE Lab (UCSD)** — research collaboration on fire-spread modeling.
- **NIST Public Safety Communications Research** — interoperability testing.

- **Copernicus Emergency Management Service** (EU).

16. Open Questions for the Agent

The agent should surface these to the human team early rather than guess:

10. Primary geography for pilot — Quebec (SOPFEU partnership) or US Pacific Northwest?
11. Bundle our own gateway hardware, or certify against existing LoRaWAN gateways?
12. Unit-cost ceiling for the Scout tier (drives sensor selection + radio choice).
13. Open-source the firmware? The mobile app? The models? Which layers stay proprietary?
14. Business model — hardware sale + SaaS? Data-as-a-service? Grant-funded non-profit? Public-private?
15. Camera SKU in v1 — privacy complexity vs. enormous analytic payoff?
16. Any mandated IRWIN / CAD integrations from a launch-customer agency?

17. Appendix A — Suggested Libraries & Tooling

- **Firmware:** ESP-IDF, Zephyr RTOS, TFLite-Micro, CBOR-C, LoRaMAC-node.
- **Backend:** Rust (ingest), Python (analytics), TimescaleDB, PostGIS, ChirpStack, EMQX, Apache Airflow, DuckDB.
- **Frontend:** React, React Native, MapLibre GL, deck.gl, Tailwind.
- **Geospatial:** GDAL, rasterio, xarray, pystac, rio-tiler, Titiler, QGIS, h3-py.
- **Fire-science:** `cffdrs` (R, CFS reference FWI/FBP — port to Python/Rust as needed), `firebehaviour` Python packages, USFS NFDRS2016 reference code.
- **ML:** PyTorch → ONNX → TFLite-Micro; Label Studio for labelling.
- **DevOps:** Terraform, Docker, GitHub Actions, Memfault (or homegrown OTA).

18. Appendix B — Naming & Branding

- **Product name:** PYRA (from *pyre / pyros* — Greek for fire). *Agent must verify TM/USPTO/CIPO clearance before any public launch.*
- **Tier names:** Scout, Sentinel, Vista. Gateway: Relay.
- **Tone:** tools-for-serious-work. Not gamified, not hype. Firefighters and researchers both respect understated, precise language.
- **Visual identity:** safety-orange accent, muted greys, topographic-contour motif. Avoid stock flame imagery.

19. First Instructions to the Agent

When you begin work, in this order:

17. Clarify the seven open questions in §16 with the human lead before spending engineering budget.
18. Build the power-budget spreadsheet and the sensor BOM with two candidates per slot. Share for review.
19. Stand up the Phase 1 single-node stack end-to-end on a bench — one stake, one laptop, one MQTT broker, FWI computed and graphed.
20. Do NOT build a new fire-danger model. Port or wrap `cffdrs`. If you think you've found an improvement, document it as a research hypothesis, not a product change.
21. Every output must carry provenance and uncertainty. No naked numbers.
22. Prefer open standards (SensorThings, STAC, GeoJSON, MQTT) over bespoke APIs. An agency that cannot ingest our data will not buy our platform.
23. Keep a running `DECISIONS.md` — every non-trivial design choice, who made it, and why. This becomes the audit trail.

End of brief. Iterate freely — this is v0.1.